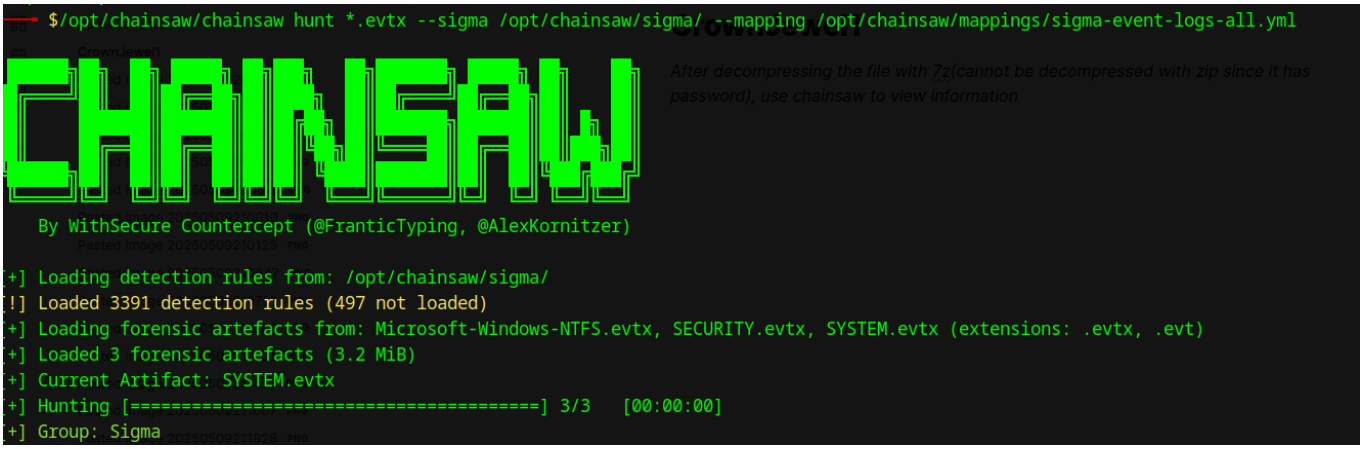
CrownJewel1

Chainsaw hunt & dump

After decompressing the file with 7z(cannot be decompressed with zip since it has password), use chainsaw to view information



It shows

1. Crash Dump
2. Scheduled task deletion
3. Rare Schtasks Creation
   Not so interesting

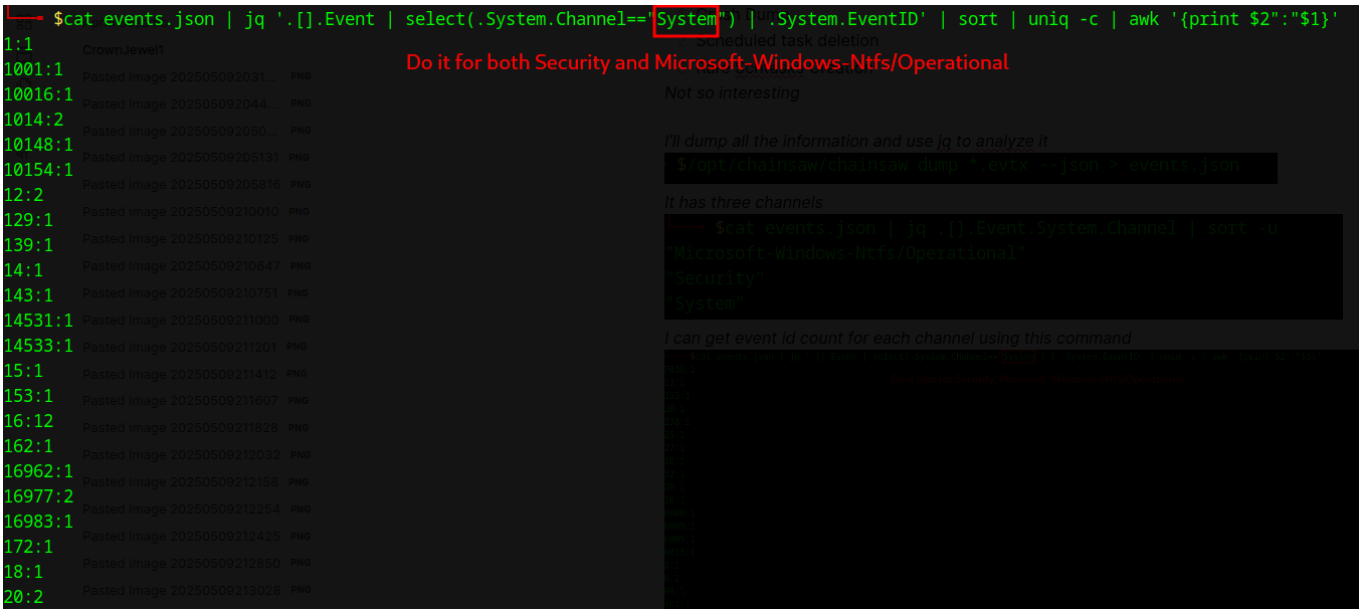
I'll dump all the information and use jq to analyze it

```
$/opt/chainsaw/chainsaw dump *.evtx --json > events.json
```

It has three channels

```
$cat events.json | jq .[].Event.System.Channel | sort -u
"Microsoft-Windows-Ntfs/Operational"
"Security"
"System"
```

I can get event id count for each channel using this command



I can use bing copilot to add information about each event id

```
1:1: System startup.
6:11: Driver loaded.
12:2: Operating system started.
18:1: Plug and Play manager started.
20:2: Device installed.
32:2: Driver installed.
41:1: The system has rebooted without cleanly shutting down first.
55:8: File system structure on the disk is corrupt and unusable.
98:2: Volume Shadow Copy Service error.
129:1: Reset to device, \Device\RaidPortX, was issued.
139:1: The system encountered a fatal error during shutdown.
143:1: The system failed to flush data to the transaction log.
153:1: The IO operation at logical block address failed due to a hardware er
162:1: The system time was updated by the BIOS.
172:1: The system detected a controller error on \Device\HarddiskX\DRX.
238:1: The disk signature of disk X is equal to the disk signature of disk \
7036:106: The service entered the running or stopped state.
7040:1: Start type of a service was changed.
6005:1: Event log service started (system boot).
6008:1: Previous system shutdown was unexpected.
6009:1: Operating system version logged at boot.
6013:1: System uptime.
```

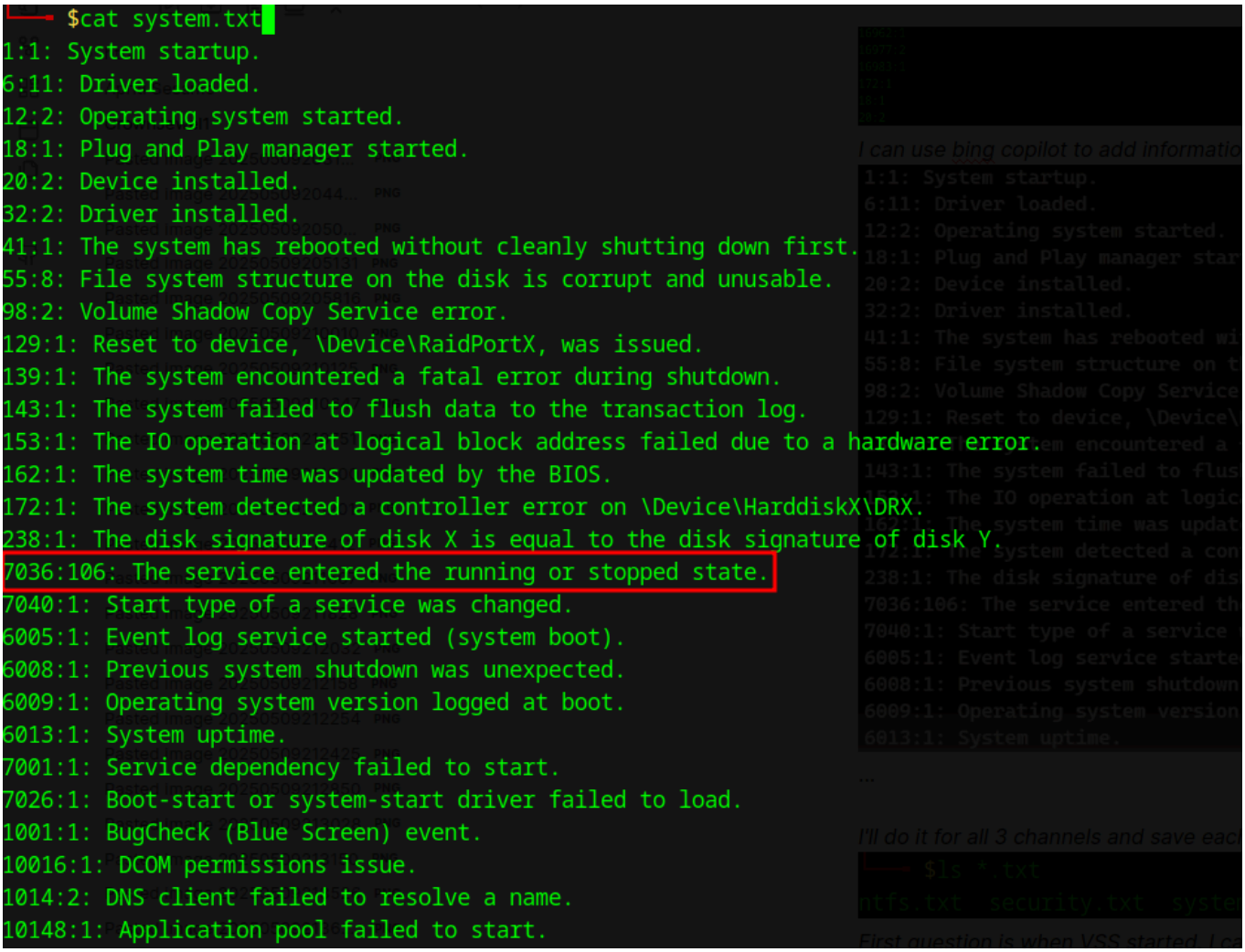
...

I'll do it for all 3 channels and save each item

```
$ls *.txt
ntfs.txt  security.txt  system.txt
```

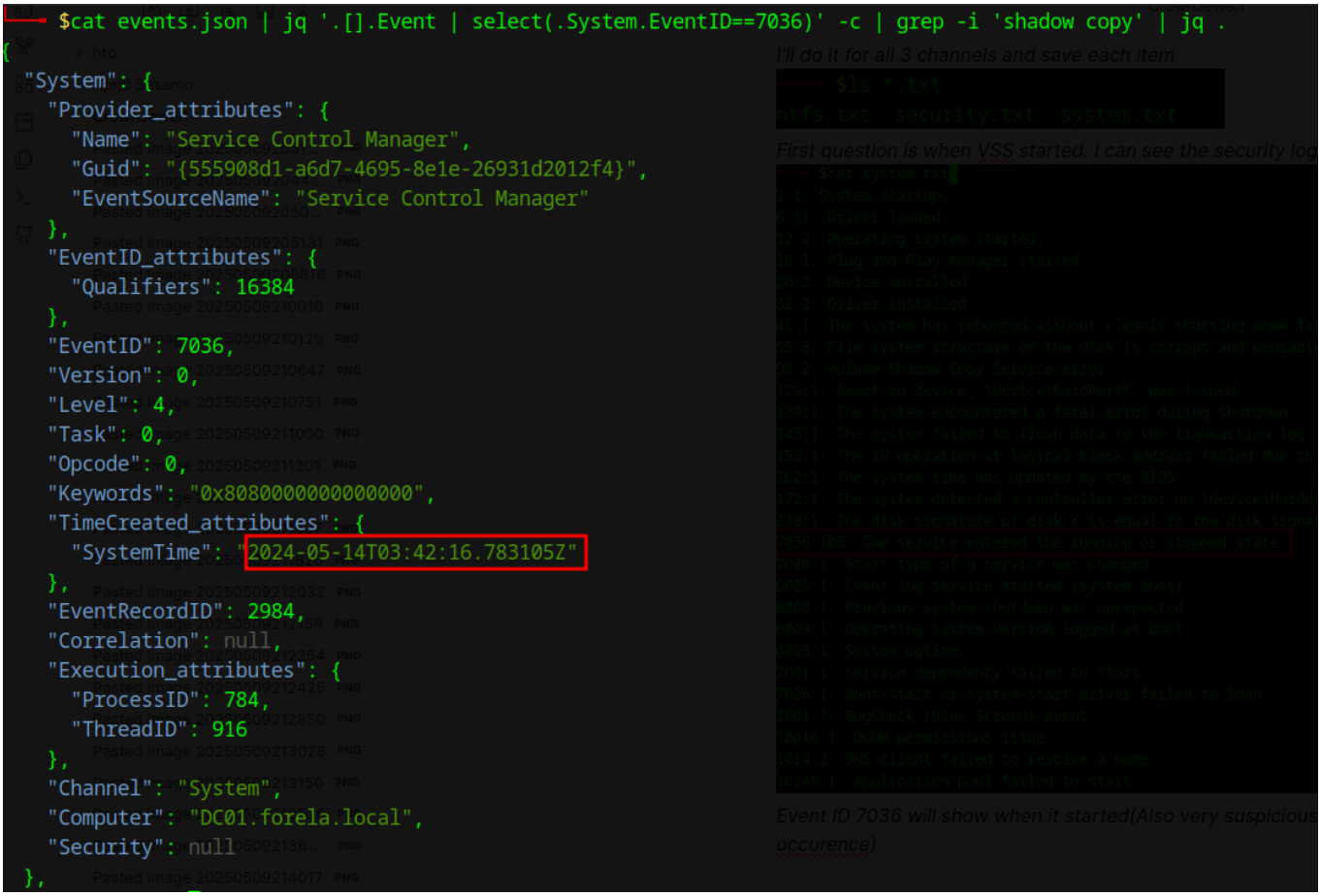
1. When VSS started

First question is when VSS started. I can see the security logs



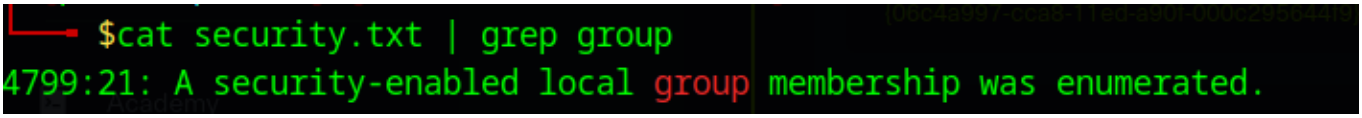
Event ID 7036 will show when it started(Also very suspicious since it has bunch of occurence)

I can see logs about when it started using jq -c and grep

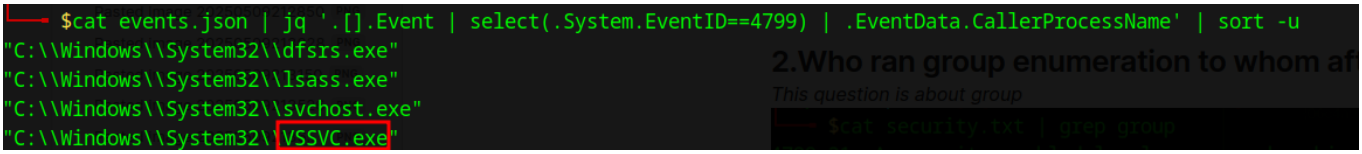


2.Who ran group enumeration to whom after VSS

This question is about group

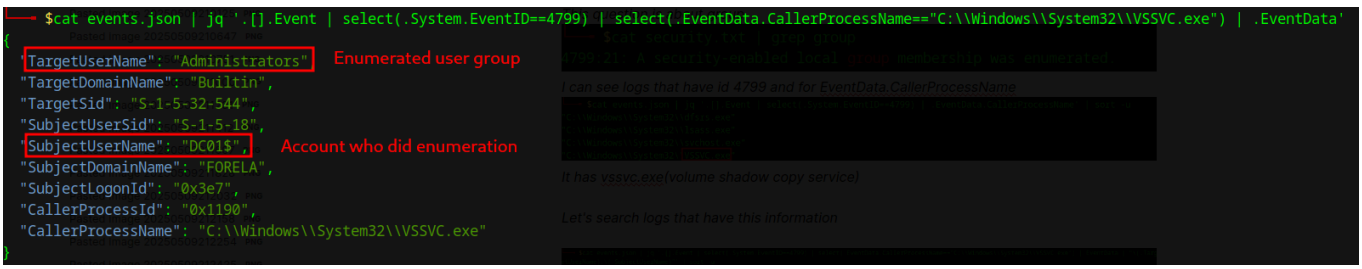


I can see logs that have id 4799 and for EventData.CallerProcessName



It has vssvc.exe(volume shadow copy service)

Let's search logs that have this information



I only want TargetUserName & SubjectUserName

```
$cat events.json | jq '.[].Event | select(.System.EventID==4799) | select(.EventData.CallerProcessName=="C:\\Windows\\System32\\VSSVC.exe") |.EventData | "\(.TargetUserName):\(.SubjectUserName)"' | sort -u
"Administrators:DC01$"
"Backup Operators:DC01$"
```

3. Process ID

It is already shown from the previous

```
$cat events.json | jq '.[].Event | select(.System.EventID==4799) | select(.EventData.CallerProcessName=="C:\\Windows\\System32\\VSSVC.exe") | .EventData'
{
  "TargetUserName": "Administrators",
  "TargetDomainName": "Builtin",
  "TargetSid": "S-1-5-32-544",
  "SubjectUserSid": "S-1-5-18",
  "SubjectUserName": "DC01$",
  "SubjectDomainName": "FORELA",
  "SubjectLogonId": "0x3e7",
  "CallerProcessId": "0x1190",
  "CallerProcessName": "C:\\Windows\\System32\\VSSVC.exe"
}
```

```
$python3 -c 'print(0x1190)'
4496
```

4. Volume ID

I can use egrep for regex filtering

```
$cat events.json | jq '.[].Event' -c | egrep -i "[0-9a-z]{8}-[0-9a-z]{4}-[0-9a-z]{4}" | grep -i shadow | jq .EventData
{
  "VolumeCorrelationId": "06C4A997-CCA8-11ED-A90F-000C295644F9",
  "VolumeIdLength": 0,
  "VolumeId": "",
  "VolumeLabelLength": 0,
  "VolumeLabel": "",
  "DeviceNameLength": 33,
  "DeviceName": "\\Device\\HarddiskVolumeShadowCopy1",
  "DeviceGuid": "00000000-0000-0000-0000-000000000000",
  "VendorIdLength": 0,
  "VendorId": "",
  "ProductIdLength": 0,
  "ProductId": "",
  "ProductRevisionLength": 0,
  "ProductRevision": "",
  "DeviceSerialNumberLength": 0,
  "DeviceSerialNumber": "",
  "BusType": 0,
  "AdapterSerialNumberLength": 0,
  "AdapterSerialNumber": "",
  "Vcb": "0xffff800f93fd41b0",
  "MountDurationUs": 0,
  "MountDuration": "0 us",
  "LongestStage": 0,
  "LongestStageDuration": "0 us",
  "LongestStagePercentage": 0,
  "SecondLongestStage": 0,
  "SecondLongestStageDuration": "0 us",
```

Regex to find ID/GUID in this format

Looks like ID for a volume

It looks like VolumeCorrelationId is the Id for the volume. Let's get only this parameter

```
$cat events.json | jq '.[].Event' -c | egrep -i "[0-9a-z]{8}-[0-9a-z]{4}-[0-9a-z]{4}" | grep -i shadow | jq .EventData.VolumeCorrelationId | sort -u
"06C4A997-CCA8-11ED-A90F-000C295644F9"
null
```

This is the only volume id that shadow copy used for

5. Copied ntds.dit full path

I can try grep ntds.dit

```
$cat events.json | jq '.[].Event' -c | grep -i ntds.dit
[✗][REDACTED][~/Downloads/Artifacts]
$
```

Nothing returns

I can dump MFT(Master File Table) data

```
$/opt/chainsaw/chainsaw dump C/\$MFT --json > MFT.json
```



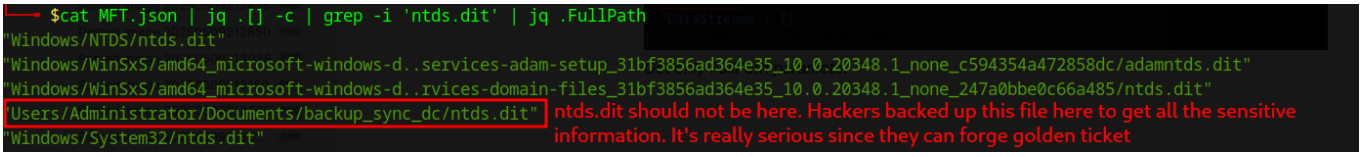

```
By WithSecure Countercept (@FranticTyping, @AlexKornitzer)

[+] Dumping the contents of forensic artefacts from: C/$MFT (extensions: *)
[+] Loaded 1 forensic artefacts (100.5 MiB)
[+] Done
```

I can grep ntds.dit

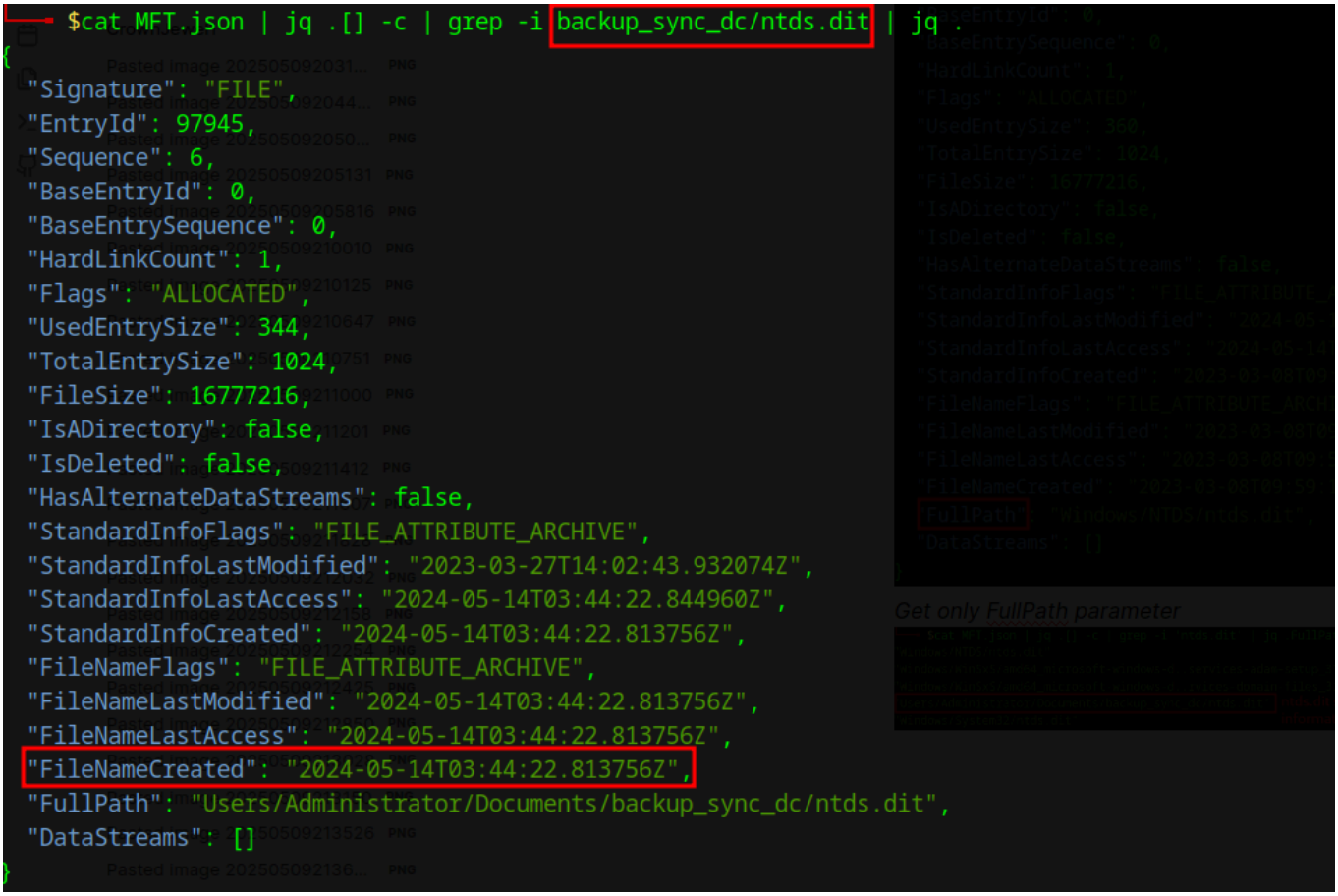
```
$cat MFT.json | jq .[] -c | grep -i 'ntds.dit' | jq .
{
  "Signature": "FILE",
  "EntryId": 697,
  "Sequence": 3,
  "BaseEntryId": 0,
  "BaseEntrySequence": 0,
  "HardLinkCount": 1,
  "Flags": "ALLOCATED",
  "UsedEntrySize": 360,
  "TotalEntrySize": 1024,
  "FileSize": 16777216,
  "IsADirectory": false,
  "IsDeleted": false,
  "HasAlternateDataStreams": false,
  "StandardInfoFlags": "FILE_ATTRIBUTE_ARCHIVE",
  "StandardInfoLastModified": "2024-05-14T03:41:53.653016Z",
  "StandardInfoLastAccess": "2024-05-14T03:41:53.653016Z",
  "StandardInfoCreated": "2023-03-08T09:59:15.917087Z",
  "FileNameFlags": "FILE_ATTRIBUTE_ARCHIVE",
  "FileNameLastModified": "2023-03-08T09:59:15.917087Z",
  "FileNameLastAccess": "2023-03-08T09:59:15.917087Z",
  "FileNameCreated": "2023-03-08T09:59:15.917087Z",
  "FullPath": "Windows/NTDS/ntds.dit",
  "DataStreams": []
}
```

Get only FullPath parameter



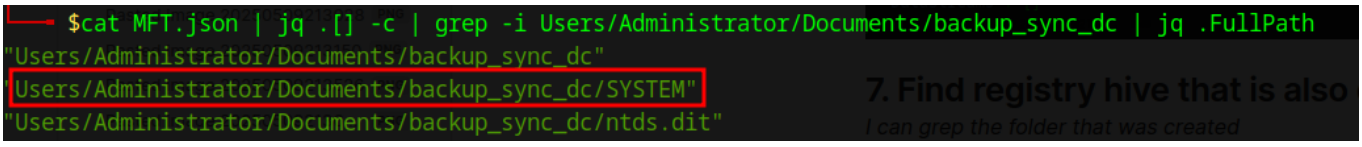
6. When this file was created

I can grep the path and search logs

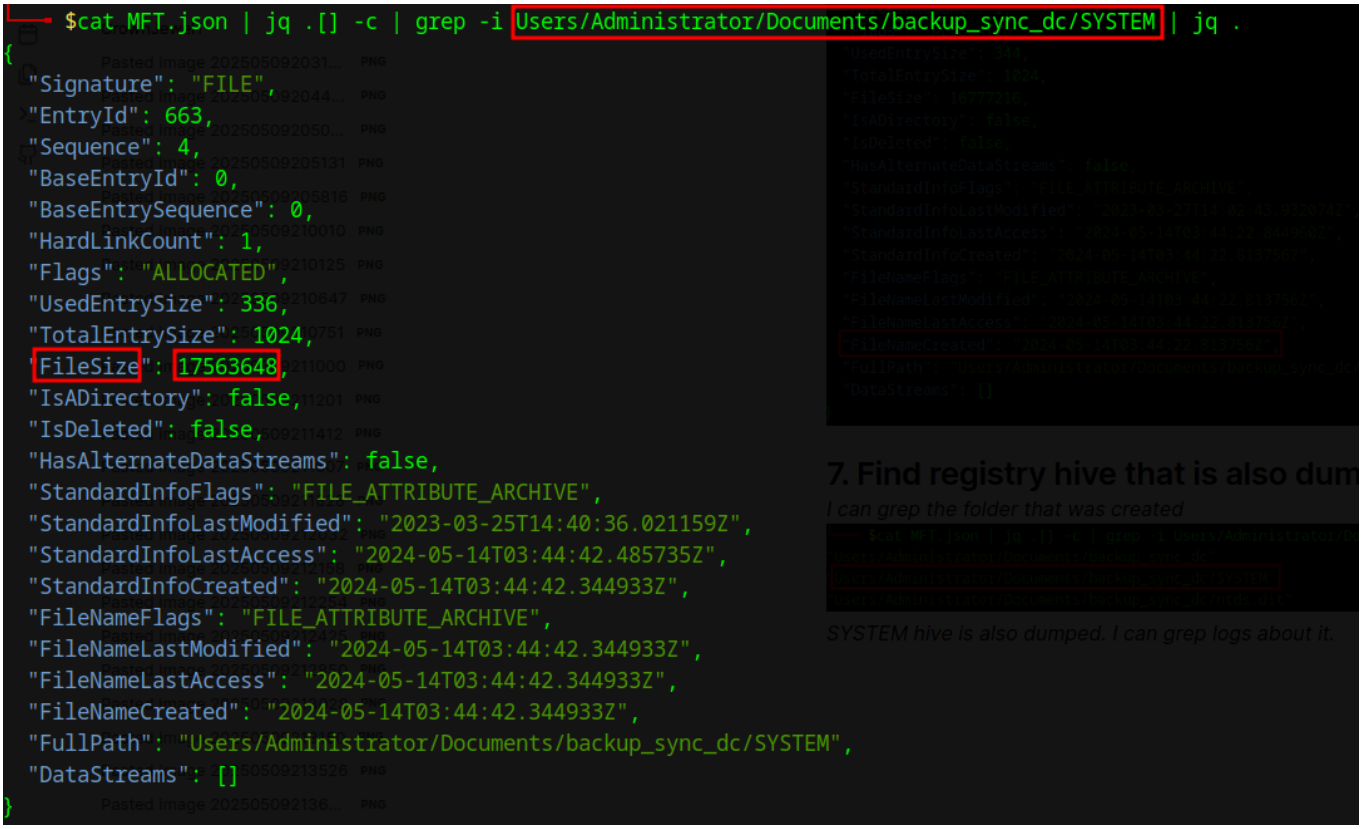


7. Find registry hive that is also dumped and its size

I can grep the folder that was created



SYSTEM hive is also dumped. I can grep logs about it.



I can also find its file size